

Week 3: Contradiction

Prove that $\frac{x^2}{y^2} = 1$ has no solutions with $x, y \in \mathbb{Z}_{>0}$.

Proof: Proceed by contradiction. Assume that $\neg P$.

Then exist $x, y \in \mathbb{Z}_{>0}$ s.t. $\frac{x^2}{y^2} = 1$.

Then $(x-y)(x+y) = 1$. Since the only

divisors of 1 are ± 1 , either

$x-y = 1$ and $x+y = 1$ or

$x-y = -1$ and $x+y = -1$.

Case 1: suppose that $x-y = 1$ and $x+y = 1$. Adding these equations gives $2x = 2$ and thus $x = 1$. But then $y = 0$. This is incompatible with our assumption that $y \in \mathbb{Z}_{>0}$. Thus case 1 can't happen.

Case 2: suppose $x-y = -1$ and $x+y = -1$.

Adding gives $2x = -2$, thus $x = -1$.

This contradicts $x \in \mathbb{Z}_{>0}$.

We conclude that $\frac{x^2}{y^2} = 1$ has no positive integer solutions. $\square$

Proof: Proceed by contradiction.

Assume $\neg P$, i.e., assume $\sqrt{2} \in \mathbb{Q}$.

By the definition of rational, $\exists a, b \in \mathbb{Z}$

s.t. $b \neq 0$ and $\sqrt{2} = a/b$. We may assume

that a and b have no common prime

factors. In particular, a and b are not

both even. Then, $a = b\sqrt{2}$. Squaring gives

$a^2 = b^2 \cdot 2$. Since the RHS is even, $2 \mid a^2$.

Since $2 \mid a^2$, 2 also divides a .

(a odd $\Rightarrow a^2$ odd)

Thus, $4 \mid a^2$, thus $4 \mid b^2 \cdot 2$.

thus $2 \mid b^2$ and so $2 \mid b$.

This contradicts a and b have no common

factors. We conclude that $\sqrt{2} \notin \mathbb{Q}$. $\square$

$$4 \mid b^2 \cdot 2 \Rightarrow 2 \mid b$$

(I) look at factorizations

$$b = 2^{e_1} \cdot 3^{e_2} \cdots$$

$$a = 2^{a_{e_1}} \cdot 3^{a_{e_2}} \cdots$$

$$b^2 \cdot 2 = 2^{2e_1+1} \cdot 3^{2e_2} \cdots$$

(II) if $2 \mid b$, then b^2 is odd

$$b^2 \cdot 2 = 2^m \Rightarrow$$

$$b^2 = 2^{m-1}$$

$$\text{odd} = \text{even}$$

Euclid's proof: Proceed by contradiction.

Suppose that there are only finitely many

primes. Label them as $p_1, p_2, \dots, p_r$.

$p_1 = 2, p_2 = 3, p_3 = 5, \dots$

Let $N = p_1 \cdot p_2 \cdots p_r + 1$

what are the factors of N ?

Let q be a prime factor of N .

$\exists i$ s.t. $q = p_i$.

Then $q \mid N$ and $q \mid p_1 \cdots p_r$, $\Rightarrow$

by 2 out of 3 rule, $q \mid 1$. $\times$

Proof: if $x+y > 5$ then $x > 2$ or $y > 3$.

Proceed by contradiction. Assume $x+y > 5$.

Assume $x \leq 2$ and $y \leq 3$,

Thus $x+y \leq 5$. This contradicts

$x+y > 5$. Thus " $x \leq 2$ and $y \leq 3$ "

was actually false. We conclude that

$x > 2$ or $y > 3$.